

Lecture 9: Applied Regression in Economics

Credibility, Diagnostics, and Knowing When to Worry

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Where we are

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Today's question

How do you know whether to believe a regression result?

Open question — vote before we start

The twins puzzle

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We will answer this precisely by the end of class.

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- ⑥ Write a 3-sentence brief that a non-economist can act on.

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Why synthetic?

In real data you never know the true coefficient.

Today we *do* know it. That's the whole point.

Homework: replicate on real CPS data.

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In applied work, the hard part is not running OLS. It is knowing which of these you have.

Interpretation: always start with units

Exercise — before you run anything

You are about to estimate:

$$EmploymentRate_i = \beta_0 + \beta_1 MinimumWage_i + u_i$$

What units is β_1 in? What would a “large” vs. “small” estimate look like?

Discuss with your neighbor for 90 seconds.

The answer depends entirely on:

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Rule: write the units of every coefficient before interpreting the number.

Functional forms — choose, don't guess

Form	Model	β_1 means ...
Level–Level	$Y = \beta_0 + \beta_1 X$	ΔY per 1 unit ΔX
Log–Level	$\ln Y = \beta_0 + \beta_1 X$	$\approx 100\beta_1\%$ ΔY per 1 unit ΔX
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More precise log rule

$$\% \Delta Y = 100 \cdot (e^{\beta_1} - 1)$$

Matters when $|\beta_1| > 0.1$

Omitted Variable Bias — with a number

Suppose the true model is:

$$Wage_i = \beta_0 + \beta_1 Educ_i + \beta_2 Ability_i + u_i$$

You can observe *Educ*, but not *Ability*. You estimate:

$$Wage_i = \alpha_0 + \alpha_1 Educ_i + v_i$$

OVB formula:

$$\text{plim } \hat{\alpha}_1 = \beta_1 + \underbrace{\beta_2 \cdot \delta_1}_{\text{bias}}$$

where δ_1 = coefficient from regressing *Ability* on *Educ*.

	Sign of β_2	Sign of δ_1
Upward bias	+	+
Downward bias	+	—

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The coefficients do not change.

The inference might.

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Today's exercise: compare a case where robust SE changes a conclusion vs. where it does not. They are not always different — knowing when it matters is the skill.

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Robust SE in Python

```
import statsmodels.formula.api as smf

# Conventional SE
m = smf.ols("log_wage ~ educ + exper + union + female", data=df).fit()

# Robust SE (HC1 -- standard in labor economics)
m_rob = smf.ols("log_wage ~ educ + exper + union + female", data=df).fit(cov_type="HC1")

# Same coefficients, potentially different SE / p-values
print(m.params.compare(m_rob.params)) # coefficients: identical
print(m.bse.compare(m_rob.bse)) # standard errors: may differ
```

Rule of thumb: if the robust SE is noticeably larger, your original inference was overconfident.

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We will build this plot in class today for the education coefficient.

Today's three-act workflow

Act	What you do	Goal
I	EDA + commit to a model	Form expectations; justify choices
II	Break the regression	Discover where it fails and why
III	Report honestly	Write for a practitioner audience

At each act, you will update a **living executive summary** — one sentence that evolves as you learn more. By the end you will have a defensible 3-sentence brief.

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Stop and write (2 minutes)

In your notebook, write your prior:

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Act II — Break the regression

Exercise A: Subsample stability

Run your preferred model on:

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Exercise B: OVB vote

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Then run it. Were you right?

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Goal: build the habit of checking before you report.

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Example

An additional year of schooling is associated with approximately a 7% increase in wages in this sample, holding experience and union status constant (robust 95% CI: 6.2–7.8%). This estimate should be interpreted as associational, not causal, as unobserved factors like ability are correlated with both education and wages. Were we able to observe ability directly, the estimate drops to around 5%, suggesting meaningful upward bias in the baseline result.

Answering the opening question

The twins puzzle — revisited

OLS of log wages on education: $\hat{\beta}_1 \approx 8\text{--}10\%$

Identical twins studies: $\approx 4\text{--}6\%$

We can now be precise:

$$\hat{\alpha}_1^{\text{OLS}} = \underbrace{\beta_1^{\text{causal}}}_{\approx 5\%} + \underbrace{\hat{\beta}_2^{\text{ability}} \cdot \hat{\delta}_1^{\text{ability on educ}}}_{\approx 3\text{--}5\%}$$

Identical twins share genetic endowment, so differencing within pairs removes most of the ability confound — that is what reduces the estimate.

This is not a curiosity. It is the central challenge of empirical economics.

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- **3-sentence executive summary** (final version).

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Homework

Replicate today's log-wage regression on real CPS microdata (IPUMS extract).

Do the planted coefficients hold?

Post your stability plot to GitHub.